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Field Trial of Intelligent Completion Technology for Horizontal Wells in Bottom - Water Reservoirs: A Comprehensive Evaluation of Productivity Improvement Strategies

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Abstract

Reservoirs with edge and bottom water are a crucial reservoir type in oilfield development. Influenced by factors including the structural height of the developed oil layers, the location of edge and bottom water layers, and oil - well production systems, during the production process, edge and bottom water rise and bottom - water coning frequently occurs. These phenomena cause premature water breakthrough and even water flooding in oil wells, significantly impeding the exertion of well productivity, reducing reservoir recovery efficiency and development benefits. Therefore, it is of great urgency to develop and apply suitable technical solutions for intervention and management, which will contribute to optimizing the development performance of such reservoirs.

This paper draws on the concept of proactive prevention and control in intelligent completion. Leveraging the advancements and development of current distributed fiber optic monitoring technology, a novel intelligent completion technology is proposed. This technology features real-time full-wellbore monitoring through distributed fiber optic cables deployed outside the casing, and production control of pay zones via switchable casing sliding sleeves. Compared with existing mainstream intelligent completion technologies, it has technical advantages such as fewer wellbore-matched tools, no impact on the inner diameter of the wellbore, and no interference with subsequent operations, thereby enhancing the technical adaptability.

A field trial was carried out, and dynamic monitoring during the stage fracturing and reconstruction process was completed. Based on the continuous real-time dynamic monitoring of the temperature by the distributed optical fiber outside the casing and the acoustic vibration, it can reflect in real time the situation of the fracturing fluid and proppant in the formation of each layer section. After the well was put into production, the daily liquid production was 10.76 m³, with a water cut of 83.5%, and the daily oil production was 1.78 tons. However, after a period of production, the bottom water coned in and broke through, causing the water cut of the well to rise to 100%. According to the distributed optical fiber monitoring, the bottom water of the reservoir was mainly observed in the third- and fourth-layer sections. The casing sliding sleeves of the third and fourth production layer sections were closed. After this measure, both the daily liquid

production and the water cut of the well significantly decreased. The daily oil production increased from 0.56 tons to 2.87 tons, achieving the technological goal of quickly restoring the productivity of the water-flooded horizontal well in the development of the reservoir with bottom water.

In response to the production contradictions in bottom water reservoirs, drawing on the ideas of existing intelligent completion technology, a novel intelligent completion string, with distributed optical fibers outside the casing and switchable casing sliding sleeves as its core, has been innovatively designed, expanding the connotation of intelligent completion technology. The field trial of intelligent completion in Well Chi* has been successful, indicating that the intelligent completion technology can achieve the monitoring and interpretation of the whole process from fracturing and reconstruction to later production of oil wells. It provides an important basis for the optimization and adjustment of later production as well as the control of production intervals. By means of comprehensive monitoring and regulation, it has effectively restored the productivity of water-flooded wells in bottom water reservoirs, which is of great significance for the efficient development and production practice of oilfields.

Introduction

Premature water breakthrough is a key challenge for oil wells in edge and bottom water reservoirs. To address this issue, operators and oilfield service companies worldwide have launched extensive technical research and field trials in recent years. These efforts fall into two main categories: passive mitigation and active prevention.

Passive mitigation strategies primarily involve remedial treatments implemented after water breakthrough initiation. Key techniques include:

- Chemical conformance control using gel-based and nano-enhanced high-strength plugging agents targeting bottom water channels (Wu, L., 2022.)
- Installation of chemical soft partitions to form impermeable barriers against bottom water encroachment (Liu, L.N., 2023, Wu, L., 2022.)
- Gas cap drive optimization in bottom water reservoirs to establish pressure gradients inhibiting vertical water coning (Liu, L.N., 2023, Wu, L., 2022.)
- Deployment of adaptive inflow control devices (AICDs) and production regime optimization to retard rapid water breakthrough (Guo, Y.F., 2017.)

While preliminary field trials have demonstrated marginal efficacy, these technologies remain in experimental application phases. Their performance is highly contingent on reservoir heterogeneities, completion configurations, and wellbore conditions, precluding widespread commercialization due to insufficient technical maturity.

Active prevention methodologies, conversely, integrate intelligent well completion principles to enable proactive water management. This approach emphasizes:

- Development of proprietary downhole toolsets
- Integration of real-time monitoring systems and zonal flow control mechanisms within the wellbore architecture
- Implementation of algorithms for interpreting production fluid characteristics to identify water entry points
- Application of stepless flow regulation across producing intervals to maintain oil stability while constraining water production

Technical surveys indicate active prevention systems for edge and bottom water reservoirs rely on integrated downhole monitoring and control architectures. Notable examples include:

- Remote-controlled completion interval isolation technologies
- Hydraulic-based stepless flow regulation systems, as validated in the Liaohe Oilfield's L X well (Peng, S.J., 2015, Wang, W., 2016, Zhang, W.P., 2022)
- Baker Hughes' MultiNode™ all-electric intelligent completion system, featuring TEC cable connectivity to surface control units paired with autonomous flow control (AFC) valves (Cui, X.J., 2021, Dang, W.H., 2016, Dou, N.H., 2018.)
- SmartWell completion technology deployed in the Penglai Oilfield (Che, Z.A., 2017, Liao, C.L., 2017, Xu, B.C. 2016.)

However, active systems face major engineering challenges: their complex mechanical design requires strict wellbore conditions, and performance degrades under paraffin deposition, scaling, or sand production. They also create constraints for future well intervention work.

Building on existing intelligent completion technology, this study designs a **novel intelligent well completion string**. Its core components are distributed optical fibers (installed outside the casing) and switchable casing sliding sleeves—expanding the scope of intelligent completion technology.

String design of intelligent well completion

The intelligent completion string has been designed through the construction of a downhole monitoring system and a production zone fluid flow control system for horizontal wells in bottom-water reservoirs, as illustrated in Fig.1.

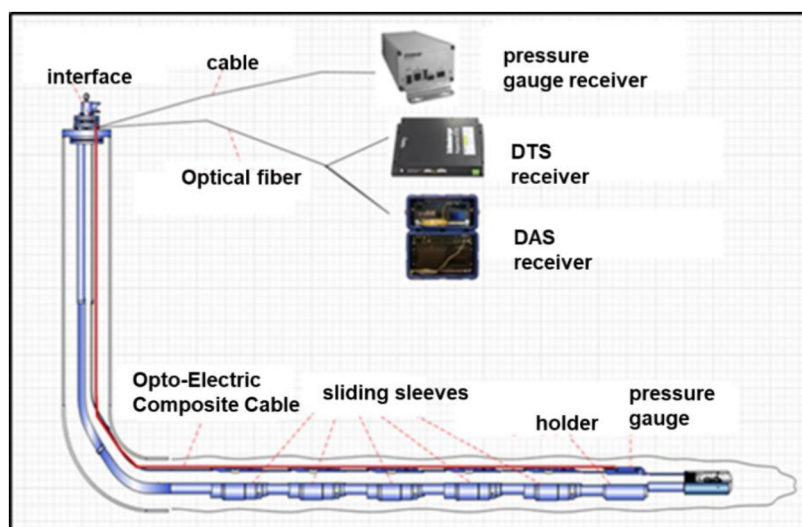


Figure 1—Diagram of Intelligent Well Completion String with External Casing Optical Fiber and Switchable Sliding Sleeve for Bottom Water Reservoirs

Monitoring System

System Configuration. Distributed optical fibers are deployed externally to the casing string, complemented by a permanent downhole pressure gauge installed at the wellbore toe. This configuration eliminates the need for internal wellbore cabling or dedicated sensor installations, preserving the integrity of the wellbore environment.

Operational Principle. The optical fiber network functions as both sensing element and data transmission medium, enabling continuous real-time acquisition of:

- Temperature profiles along the entire wellbore, including all production intervals

- Acoustic vibration signatures associated with fluid flow dynamics
- Downhole pressure data from the permanent gauge

These multi-parametric datasets are integrated to interpret fluid flow events at discrete wellbore locations, facilitating comprehensive characterization of production behavior across reservoir intervals and identification of abnormal points (e.g., casing leaks due to mechanical failure).

Technical Advantages. This design offers three key benefits:

- Unaltered internal wellbore conditions, supporting subsequent intervention operations and auxiliary tool deployment
- Simplified system architecture with reduced component complexity
- Enhanced adaptability to harsh downhole environments (e.g., scaling, wax deposition) compared to conventional cabled systems

Control System

Hardware Configuration. Production zone flow control is achieved through strategically placed switchable casing sliding sleeves, which are run in-hole as an integral component of the completion string during primary well construction.

Operational Sequence.

Fracturing Phase

Coiled tubing-deployed key tools sequentially actuate the sliding sleeves to open positions, enabling targeted stimulation of individual reservoir intervals.

Production Phase

Real-time data from the optical fiber monitoring network and downhole pressure gauge are used to:

- Interpret fluid composition and flow characteristics across production zones
- Identify water breakthrough intervals through signature analysis

Remedial Phase

Selective closure of sliding sleeves in identified water-producing zones is performed via wellbore intervention, diverting production to remaining oil-bearing intervals.

Performance Objectives. This control mechanism achieves two primary technical goals:

- Precise isolation of water-producing intervals to maintain oil production stability
- Rapid optimization of reservoir drainage efficiency by prioritizing productive zones

Design of key tools

To implement the intelligent completion technology featuring distributed optical fibers outside casing and switchable casing sliding sleeves for bottom water reservoirs, a full set of compatible tools has been designed and fabricated.

Optical Fibers Externally Mounted on Casing

Optical fibers externally mounted on casing are primarily used to monitor temperature and acoustic vibration signals at different positions throughout the wellbore. Thus, a three-core optical fiber configuration is adopted, as illustrated in Fig.2 (a): 1 single-mode fiber for acoustic vibration monitoring and 2 multi-mode

fibers for temperature profile monitoring. Additionally, to monitor bottom-hole pressure and provide a basis for interpreting production status, a casing pressure gauge (Fig.2 (b)) and a holder (Fig.2 (c)) are required. Considering the immature technology and high cost of fiber optic pressure gauges, a mature electronic pressure gauge is selected. Therefore, the optical fibers externally mounted on casing are ultimately designed as an opto-electric composite cable for process matching. The opto-electric composite cable features a corrosion-resistant 825 alloy steel tube inner armor and a square PVC outer armor. Functionally, it meets the comprehensive technical requirements of optical fiber temperature and acoustic monitoring, as well as power supply and data transmission for the electronic pressure gauge. It also satisfies the wear resistance, pressure resistance, and tensile performance during the simultaneous running with the casing, thereby ensuring the successful permanent external installation of the opto-electric composite cable in casing completion.

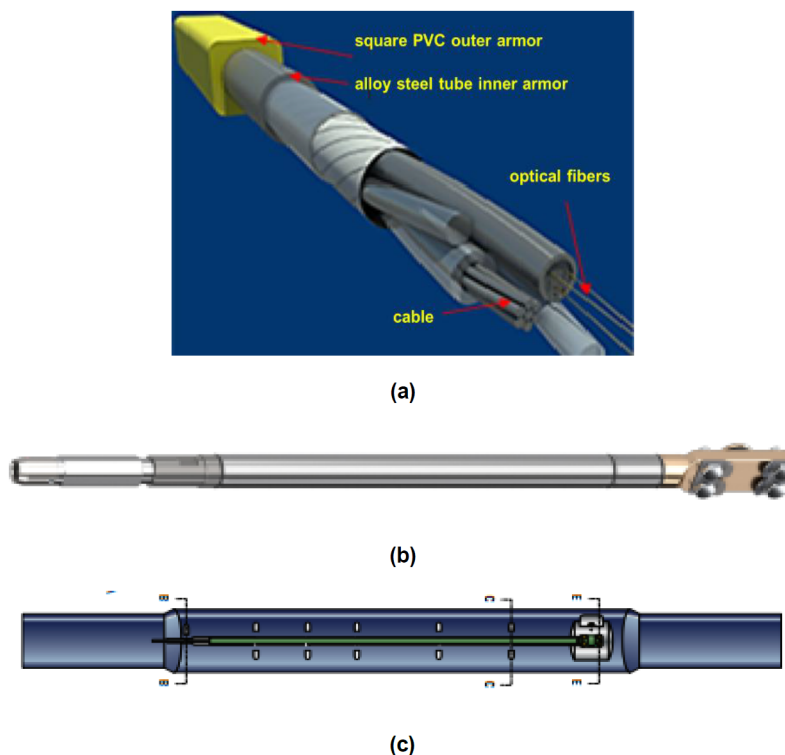


Figure 2—Schematic Diagram (a) Opto-Electric Composite Cable, (b) Casing Pressure Gauge, and (c) holder

Switchable Casing Sliding Sleeve

A new type of switchable casing sliding sleeve (Fig.3 (a)) made of P110 material has been designed and fabricated. Its internal diameter is the same as that of the completion casing, ensuring no impact on subsequent wellbore treatment and stimulation operations. A cable-passing groove is arranged on the outside of the sliding sleeve for protection and traversal of the opto-electric composite cable during running. The sliding sleeve switch is designed with rubber ring sealing, featuring a pressure-bearing capacity of 70 MPa and a temperature resistance of 120°C, which meets the requirements of pressure sealing during reservoir stimulation fracturing and sealing performance during subsequent zone closing. The opening force of the sliding sleeve is designed to be 1.5-2.5 t, ensuring small pulling force for opening and closing operations and high success rate. The sliding sleeve switching tool (Fig.3 (b)) is designed with a hydraulically driven, convex key structure. When the internal driving pressure of the tool is low, the convex keys can be automatically retracted, resulting in an overall simple structure and high safety.

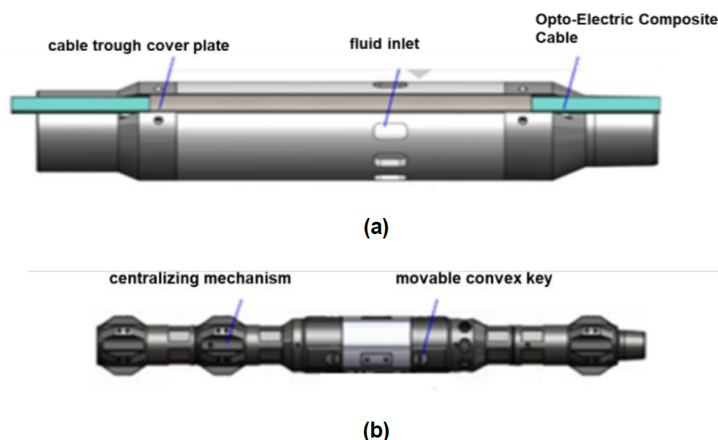


Figure 3—3D Structural Diagram and (a) Switchable Casing Sliding Sleeve; (b) Switching Tool.

Fittings

To ensure the smooth deployment of the intelligent completion technology in the field, supporting tools have been designed and fabricated, mainly including an opto-electric composite cable traversing casing head (Fig.4 (a)), a wellhead composite cable sealer (Fig.4 (b)), composite cable protection clips (Fig.4 (c)), and a surface control cabinet (Fig.4 (d)). The composite cable traversing casing head realizes the wellhead traversal and sealing of the composite cable through a traversing hole set on the casing head, with a pressure resistance of 70 MPa. The wellhead composite cable sealer mainly realizes the separate traversal and sealing of optical cables and electrical cables after the composite cable traverses the casing head, with a traversal sealing pressure resistance of 103 MPa, ensuring more stable signal transmission. Special casing composite cable protection clips are designed, with cable grooves inside the clips. The effective protection of the composite cable is achieved through the clip skeleton and cable groove fixing. After the opto-electric composite cable traverses the wellhead, it is connected to the surface control cabinet, where optical cable and electrical cable connectors are installed and protected, and connected to optical cable and electrical cable acquisition devices and a portable computer, realizing continuous real-time monitoring, storage, and display of downhole signals.

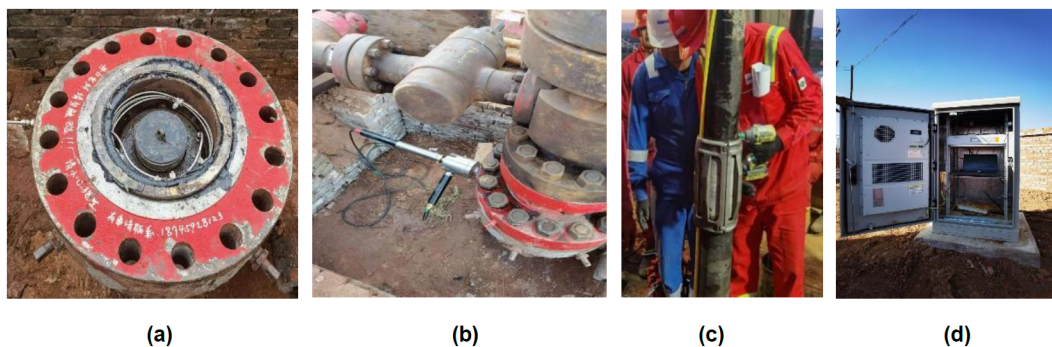


Figure 4—Pictures of Fittings: (a) casing head traversal; (b) wellhead sealer; (c) composite cable protector; (d) surface control cabinet.

Field trial

To evaluate the performance of the process system and fittings, a field trial was conducted on Well C*, enabling dynamic monitoring throughout the entire process from fracturing stimulation to subsequent production. Based on the monitoring results, the well's productivity was effectively restored through sliding sleeve closing operations.

Well Condition

Well C* develops the Triassic Chang 2 oil layer with developed bottom water. The effective thickness is 6.8 m, resistivity is 8.0 Ω m, acoustic travel time is 235.0 μ s/m, porosity is 12.9%, and apparent permeability is 3.4 mD. The total depth of the well is 2493 m, with a horizontal section length of 173 m, and it is designed to be fractured in 5 stages. In October 2022, an intelligent completion test was carried out on this well. The process supporting tools such as the opto-electric composite cable externally mounted on the casing, switchable sliding sleeves, downhole permanent pressure gauge, and support sleeve were successfully run to the designed positions, and downhole dynamic monitoring and interpretation from fracturing stimulation to subsequent production were completed smoothly.

Monitoring Results and Interpretation Analysis

Dynamic monitoring during the fracturing stimulation of the intervals was completed. Based on the continuous real-time dynamic monitoring of temperature and acoustic vibration by the distributed optical fiber outside the casing, the entry of fracturing fluid and proppant into the formation in the intervals could be reflected in real-time (Fig. 5).

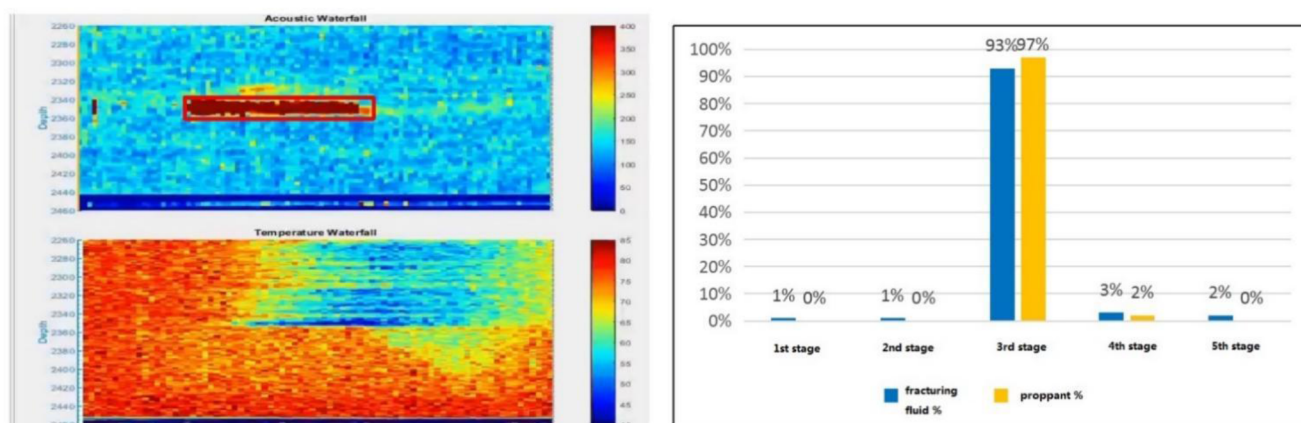


Figure 5—Optical Fiber Monitoring Result of Fracturing Stimulation in the Third Interval

The monitoring of the full wellbore temperature profile during the shut-in and production stages after completion showed significant temperature changes, mainly caused by the production from the pay zones and fluid flow in the wellbore (Fig. 6).

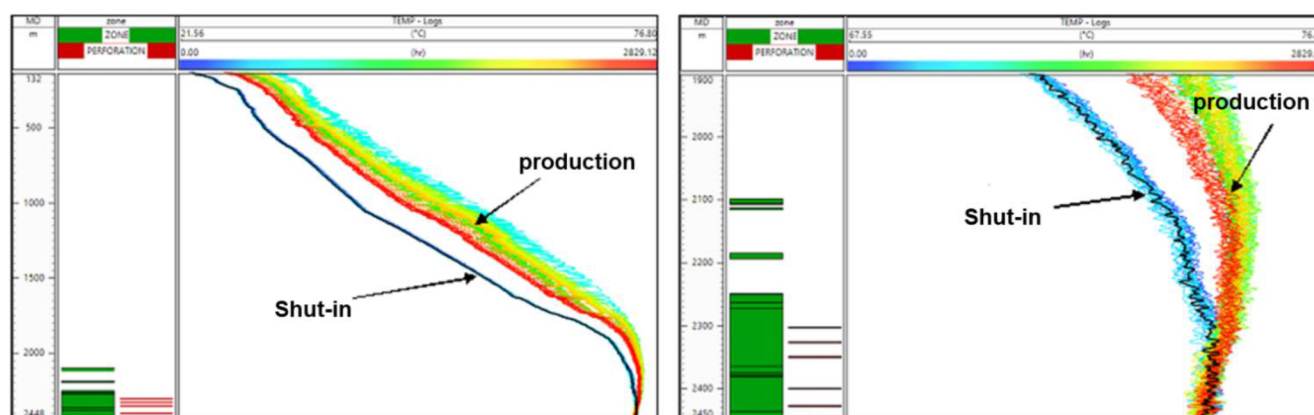


Figure 6—Optical Fiber Monitoring of Full Wellbore Temperature Profile Monitoring

After putting into production, the daily fluid production was 10.76 m³, with a water cut of 83.5% and a daily oil production of 1.78 tons. However, after a period of production, bottom water coning breakthrough occurred, resulting in the water cut rising to 100% (Fig. 7). According to the distributed optical fiber monitoring, the bottom water breakthrough mainly occurred in the third and fourth intervals.

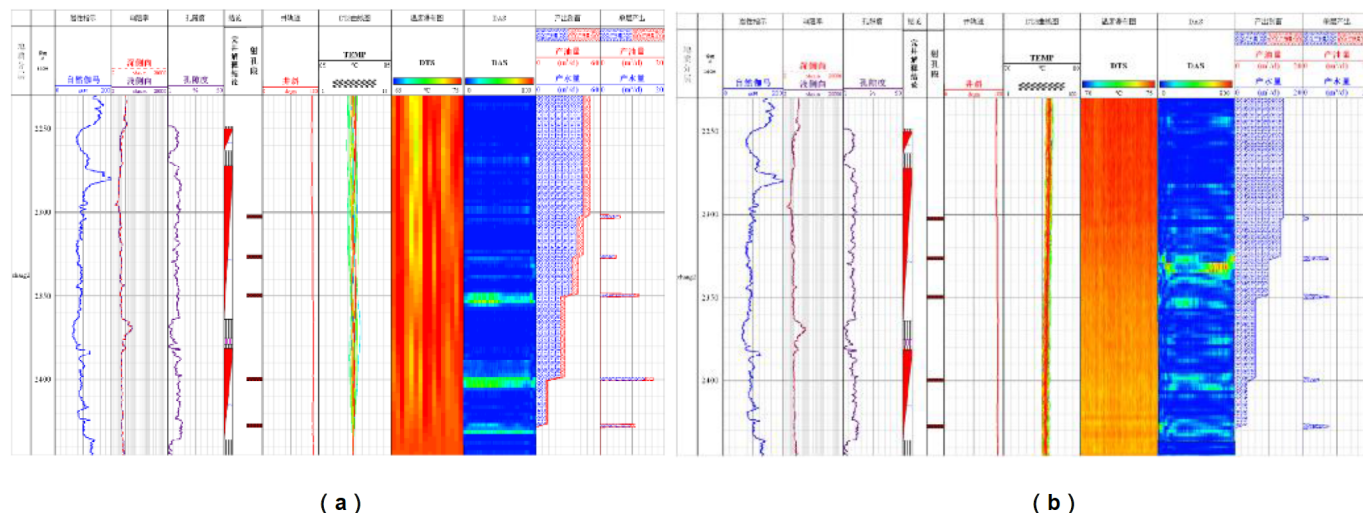


Figure 7—Interpretation Diagram of Fluid Production Profile Monitoring Results: (a)in the Early Stage of Production;(b)in the Late Stage of Production.

Production Interval Regulation Operations and Results

In response to the water breakthrough after the completion and production of Well C*, to effectively restore the well's productivity, optimization and adjustment of the production system and operations to close the casing sliding sleeves in the water-producing intervals were carried out. Initially, the production system of the well was optimized (mainly by reducing production parameters) to further reduce the bottom-hole production pressure difference and alleviate the degree of bottom water coning in the pay zones. From the production dynamic curve (Fig.8), a certain regulatory effect was observed, but it was limited. In the later stage, ordinary tubing operations were used to close the casing sliding sleeves in the third and fourth production intervals (Fig.9). After the measures, both the daily fluid production and water cut of the well decreased significantly, and the daily oil production increased from 0.56 tons to 2.87 tons, initially realizing the productivity restoration of the well. This indicates the success of the intelligent completion test of distributed optical fiber outside casing + switchable casing sliding sleeves in bottom water reservoirs.

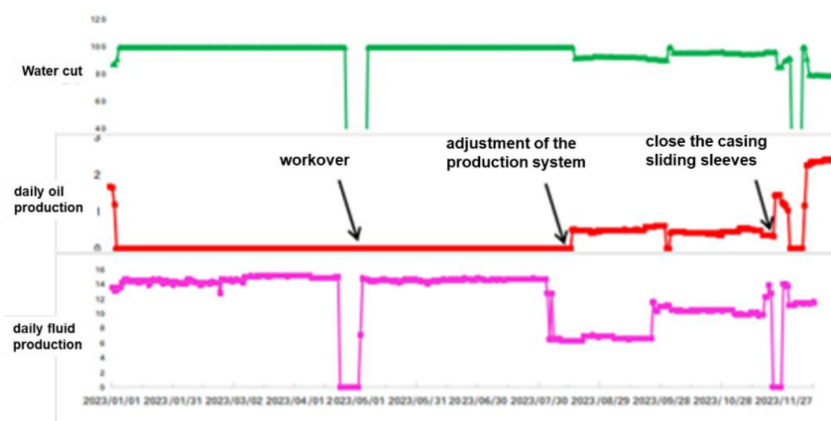


Figure 8—Production Dynamic Curves Before and After Closing the Casing Sliding Sleeves.

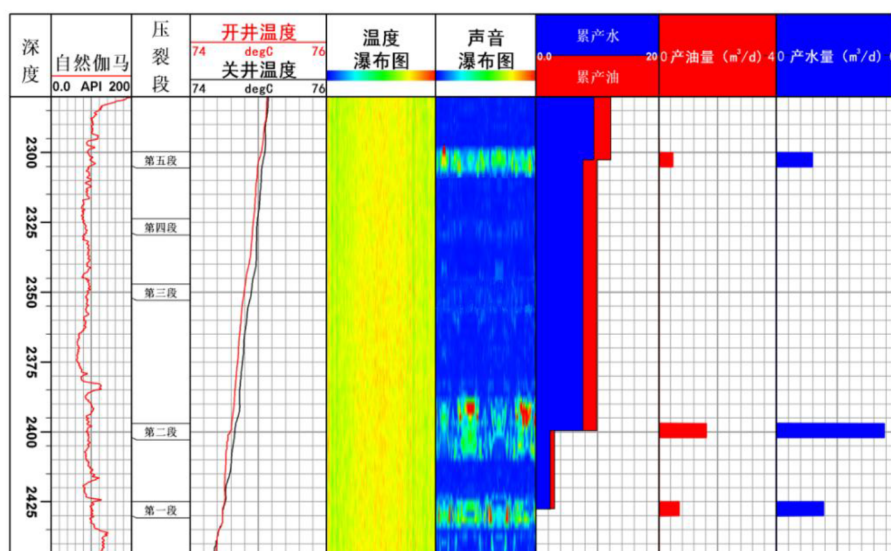


Figure 9—Optical Fiber Monitoring Fluid Production Profile Diagram After Sliding Sleeve Closing.

Conclusions

1. To solve production challenges in bottom water reservoirs, a novel intelligent well completion string was developed. It builds on existing intelligent completion technologies and centers on two core components: distributed optical fibers (outside the casing) and switchable casing sliding sleeves. This design broadens the technical scope of intelligent completion.
2. A full set of matching tools was designed for the new completion string. All key performance indicators meet the technical requirements of the process.
3. The successful field trial in Well C* confirms the value of the new technology:
 - o It enables full-process monitoring of oil wells—from fracturing stimulation to production—and supports data interpretation;
 - o It provides a basis for later production optimization, adjustments, and zone control;
 - o Through integrated monitoring and regulation, it restores productivity in water-flooded bottom water reservoir wells.

This technology offers important guidance for efficient oilfield development practices.

Nomenclature

Acoustic Travel Time	: $\mu\text{s/m}$
AFC	: Autonomous Flow Control
AICD	: Autonomous Inflow Control Device
Permeability	: mD
Pressure Resistance	: MPa (Megapascal)
Resistivity	: Ωm
TEC	: Tube Electrical Conductor

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